

Course Code	Name of the Course				
216U03E614	Cyber Security				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		--	20	30	50

Course pre-requisites: Computer networks

Course Objectives: This course provides students basic knowledge and skills in the fundamental theories and practices of Cyber Security. The course will give the knowledge and techniques of protecting computers, networks, programs and data from unauthorized access or attacks that are aimed for exploitation. The course also provides an overview on the future trends in the security domain.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the broad set of technical, social & political aspects of Cyber Security
- CO2.** Understand the nature of secure software development and operating systems
- CO3.** Discuss the importance of attacks and preventive measures at network and application layer
- CO4.** Discuss strategies for cyber defense and incident response
- CO5.** Elaborate about cyber security policies, risk management and future trends in security domain.

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction to cyber security		07	CO1
	1.1	Introduction to Cyber Security and Information Security		
	1.2	Threats , harms, vulnerabilities and controls		
	1.3	Attack Vectors (Types of Attacks, Types of Malwares – Worms, Phishing, Trojans, Botnets, Root Kits and Viruses etc),Types of security, goals of security		
	1.4	Need of Cryptography(public and private key) hashing, session keys		
2	Software and System Security		13	CO2
	2.1	Firewalls Basics, Stateless packet filtering, Stateful/Session Filtering, Application Level Firewalls, Circuit Level Gateways, Packet Filter Vs Firewall, Packet Characteristic to Filter, Linux Firewall, Windows Firewall, Network Address Translation (NAT) and Port Forwarding, Basics of Virtual Private Networks		
	2.2	Security in Operating Systems, Access Control and Authentication, Patch Management and Updates, Secure Configuration and Hardening		
	2.3	Secure Software Development Lifecycle (SDLC)		
3	Network Security & Application Security		11	CO3
	3.1	Security Issues in TCP/IP- TCP, DNS, Routing Protocol (Routing Security –BGP ,vulnerabilities, BGP Security etc)		
	3.2	Web Application Security-OWASP security and Large Language Model Applications security		
	3.3	Content Security Policies (CSP) in web – HTTP Header Directives and Sources, Session Management and User Authentication, Session Integrity, HTTPS, SSL/TLS (working principle)		
4	Cyber Threats and Defense Mechanism		08	CO4
	4.1	Denial of Service (DoS) and Distributed Denial of Service (DDoS) Attacks, Advanced Persistent Threats (APTs)		
	4.2	Cyber Defense Strategies ,Intrusion Detection System/ Intrusion Prevention System – Overview, Anomaly Based Detection Methods, Signature Based IDS/IPS, Network based IDS/IPS,		

		Host Based IDS/IPS, SNORT, Honeypot		
	4.3	Challenges ,Best Practices, Security Audits and Assessments, Limitations ,Techniques to visualize network data		
5	Cyber Security Policies, Risk Management and Future Trends		06	CO5
	5.1	Cyber Security Governance and Policies, Risk Management Frameworks, Digital and Cyber Forensics Compliance and Legal Issues in Cyber Security, Business Continuity and Disaster Recovery Planning, Privacy and Data Protection Regulations		
	5.2	Overview of Industrial Internet of Things (IIoT) Security, Cloud Security, Artificial Intelligence and Cyber Security ,Quantum Cryptography		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Nina Godbole, Sunit Belapure	<i>Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives</i>	Wiley india Pvt. Ltd, India	1st edition copyright 2011 reprint 2013
2	<u>Mike Shema</u>	<i>Anti-Hacker-Tool -Kit</i>	McgrawHill, India	4th Edition February 2014
3	Charles Pfleeger, Shari Lawrence Pfleeger, Jonathan Margulies	<i>Security in Computing</i>	Prentice Hall, India	5th Edition February 5, 2015
4	William Stallings	<i>Network Security Essentials: Applications and Standards</i>	Pearson	February ,2018
5	Chwan-Hwa (John)Wu J. David Irwin	<i>Computer Networks and Cyber Security</i>	CRC Press	First Indian Reprint, 2014